

Rule No.	Organism(s)	Indicator Agent	Agents Affected	Rule	Remarks	Grade	References
Beta-lactams							
1	Beta-Haemolytic streptococci (Group A, B, C, G)	benzylpenicillin	aminopenicillins, cephalosporins and carbapenems	IF susceptible to benzylpenicillin THEN report susceptible to aminopenicillins, cephalosporins and carbapenems. IF resistant to benzylpenicillin, THEN confirm species identification and determine and report according to the MIC of the relevant agent.	Rare isolates of group B streptococci may have diminished susceptibility to penicillins No resistance to beta-lactams reported so far except in Group B streptococci (MIC of benzylpenicillin up to 1 mg/L). If beta-haemolytic streptococci including GBS test resistant to penicillin check identification and susceptibility	C	
2	Viridans group streptococci	benzylpenicillin screening test	aminopenicillins and cefotaxime or ceftriaxone	IF susceptible in the benzylpenicillin screening test, THEN report susceptible to any indicated beta-lactam agent; IF resistant in the benzylpenicillin screening test, THEN test and report agents for which there are breakpoints, individually.	Production of mosaic PBPs leads to various patterns of beta-lactam resistance so in case of resistance to benzylpenicillin the results of other drugs cannot be inferred	C	Pottumarthy & Morris, 1998
Macrolides, lincosamides and streptogramins							
3	<i>Streptococcus spp.</i>	erythromycin, clindamycin	clindamycin	IF resistant to erythromycin AND susceptible to clindamycin THEN test for inducible MLS _B resistance IF negative THEN report clindamycin susceptible IF positive THEN report clindamycin resistant	Streptococci resistant to macrolides but susceptible to clindamycin produce Erm ribosomal methylases conferring the inducible MLS _B phenotype or express efflux pumps. In case of inducible MLS _B resistance, constitutively resistant mutants can be selected by clindamycin. Adding a note may be considered, stating that clindamycin, if testing susceptible, may still be used in less severe skin and soft tissue infections.	A	Lewis et al., 2014

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					For beta-haemolytic streptococci a note may be considered stating that clindamycin, if testing susceptible, even in the presence of inducible MLS _B resistance, may be used in combinations when reduction of toxin synthesis is desired, such as streptococcal fasciitis		
Fluoroquinolones							
4	<i>Streptococcus</i> spp. A, B, C, G	norfloxacin screening test	levofloxacin, moxifloxacin	IF susceptible in the norfloxacin screening test THEN report susceptible to levofloxacin and moxifloxacin IF resistant in the norfloxacin screening test THEN report levofloxacin and moxifloxacin resistant OR test the individual agents and report as tested	As with other gram-positive organisms, first step mutants as well as overexpressed efflux pumps are detected with norfloxacin; therefore, norfloxacin susceptible isolates can be reported as susceptible to the other fluoroquinolones.	A	Petrelli et al., 2014; Pinho, Melo-Cristino, Ramirez, & Portuguese Group for the Study of Streptococcal Infections, 2010
Aminoglycosides							
5	<i>Streptococcus</i> spp.	gentamicin	gentamicin	IF high-level resistance to gentamicin is detected THEN report with a cautionary warning that combinations of a beta-lactam with gentamicin may not be synergistic IF high level resistance to gentamicin is not detected, THEN report gentamicin as effective for synergy purposes NOTE: this rule applies to isolates from endocarditis only.	High-level resistance to aminoglycosides has been detected in viridans streptococci as well as in group B streptococci. There are in vitro data suggesting reduced synergy in such isolates; however, clinical or experimental data showing increased failure rates are lacking. Analogous to enterococci, combinations of beta-lactams with aminoglycosides in therapy should be approached with caution.	C	Farber & Yee, 1987; Kaufhold & Potgieter, 1993; Doumith et al., 2017

References

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